This document provides additional assistance with wiring your Extron IP Link Pro Control Processor to your device. Different components may require a different wiring scheme than those listed below.

For complete operating instructions, refer to the user's manual for the specific IP Link Pro Control Processor or the documentation supplied by the manufacturer of the controlled device.

For more information on using Global Scripter Modules, refer to the "Guide to Using Scripter Modules" document.

## Device Specifications

Device Type: Recorder
Manufacturer: AJA
Firmware Version: N/A
Model(s): Ki Pro Ultra

## Tested on the Following Software and Firmware Versions

| IP Link Pro Control Processor Firmware | Global Scripter Version |
|---|---|
| 3.06.0001-b002 | 2.3.1 |

## Version History

| Module Version | Date | Notes |
|---|---|---|
| 1_0_0_0 | 4/30/2020 | Initial Version |

## Module Notes

- Unidirectional variable must be set to 'True' if status is not required. Default value is 'False'.
  `Example: InterfaceName.Unidirectional = 'True'`

- connectionCounter variable must be set to the number of queries that will be sent to the device before displaying 'Disconnected' if no response is received. Default value is 15.
  `Example: InterfaceName.connectionCounter = 5`

## Supported Classes and Examples

**SerialClass**

```
    InterfaceName = ModuleName.SerialClass(ProcessorName, 'COM1', Model='Ki Pro Ultra')
```

**SerialOverEthernetClass**

```
    InterfaceName = ModuleName.SerialOverEthernetClass('192.168.254.254', 2001, Model='Ki Pro
Ultra')
```

**HTTPClass**

```
    InterfaceName = ModuleName.HTTPClass('192.168.254.254', 80, 'admin', 'password', Model='Ki Pro
Ultra')
```

## Control Commands (Serial)

Format with Qualifier:

```
InterfaceName.Set(Command, Value, {'Qualifier Key': 'Qualifier Value'})
```

Format without Qualifier:

```
InterfaceName.Set(Command, Value)
```

| Command | Value | Value | Value |
|---|---|---|---|
| **Eject** | None | | |
| `# Eject example`<br>`InterfaceName.Set('Eject', None)` | | | |
| **Standby** | 'On' | 'Off' | |
| `# Standby example`<br>`InterfaceName.Set('Standby', 'On')` | | | |
| **Transport** | 'Play'<br>'Fast Forward' | 'Stop'<br>'Rewind' | 'Record' |
| `# Transport example`<br>`InterfaceName.Set('Transport', 'Play')` | | | |

## Control Commands (Ethernet)

Format with Qualifier:

```
InterfaceName.Set(Command, Value, {'Qualifier Key': 'Qualifier Value'})
```

Format without Qualifier:

```
InterfaceName.Set(Command, Value)
```

| Command | Value | Value | Value |
|---|---|---|---|
| **Record** | 'Start' | 'Stop' | |

```
# Record example
InterfaceName.Set('Record', 'Start')
```

| Command | Value | Value | Value |
|---|---|---|---|
| **Transport** | 'Play' | 'Stop' | 'Fast Forward' |

```
# Transport example
InterfaceName.Set('Transport', 'Play')
```

## Status Available (Ethernet)

For all commands, call Update to receive the latest status. ConnectionStatus does not support the Update function and is triggered by the device providing a successful response to other Update function calls.

Format with Qualifier:

```
InterfaceName.Update(Command, {'Qualifier Key': 'Qualifier Value'})
Value = InterfaceName.ReadStatus(Command, {'Qualifier Key': 'Qualifier Value'})
InterfaceName.SubscribeStatus(Command, {'Qualifier Key': 'Qualifier Value'}, FeedbackHandler)
FeedbackHandler will be called only when the specified qualifier gets a new status.
```

Format without Qualifier:

```
InterfaceName.Update(Command)
Value = InterfaceName.ReadStatus(Command)
InterfaceName.SubscribeStatus(Command, None, FeedbackHandler)
FeedbackHandler will be called when any qualifier gets a new status.
```

| Command | Value | Value | |
|---|---|---|---|
| **ConnectionStatus** | 'Connected' | 'Disconnected' | |

```
# ConnectionStatus example
Value = InterfaceName.ReadStatus('ConnectionStatus')
InterfaceName.SubscribeStatus('ConnectionStatus', None, FeedbackHandler)
```

| Command | Value | Value | Value |
|---|---|---|---|
| **RecordingState** | 'Uninitialized' | 'Idle' | 'Recording' |
| | 'Failing In Idle' | 'Failing In Record' | 'Shutdown' |

```
# RecordingState example
InterfaceName.Update('RecordingState')
Value = InterfaceName.ReadStatus('RecordingState')
InterfaceName.SubscribeStatus('RecordingState', None, FeedbackHandler)
```

## Cable and Adapter Requirements

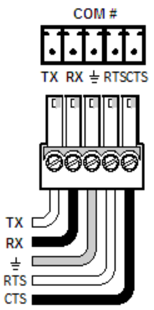
Captive Screw to Male RS422 DB9 Serial Cable

## Notes for the Device

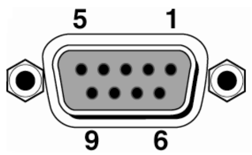
## Serial communication

| | | | |
|---|---|---|---|
| **Port Type:** | RS-422 | **Parity:** | None |
| **Baud Rate:** | 38400 | **Stop Bits:** | One |
| **Data Bits:** | 8 | **Flow Control:** | None |

## Pin Assignments Diagram




| Signal | Main Cable | Pin | Signal |
|---|---|---|---|
| Tx- | ⟶ | 8 | Rx- |
| Rx- | ⟵ | 2 | Tx- |
| GND | —— | 4 | GND |
| Tx+ | ⟶ | 3 | Rx+ |
| Rx+ | ⟵ | 7 | Tx+ |



RS422 DB9, Female

## Network communication

When configuring the Ethernet module, be sure device settings match those of the Global Scripter ethernet interface.

| | |
|---|---|
| **Port Type:** | Ethernet |
| **Default Port:** | 80 |
| **Default Username:** | |
| **Multi-Connection Capabilities:** | Undetermined |
| **Port Changeability:** | Undetermined |

## Ethernet Module Configuration Description

Please refer to user manual for settings and changes to the network communication

## Notes for the Device

- Please ensure that Authentication is Disabled for driver functionality. The following URL can be sent from a web browser, command line cURL, or Postman:
  - http://IP_ADDRESS/config?action=set&paramid=eParamID_Authentication&value=0

## Appendix A. Set Commands(Serial)

| | |
|---|---|
| **Eject None** | \x0F/ |
| **Standby Off** | \x04$ |
| **Standby On** | \x05% |
| **Transport Fast Forward** | \x100 |
| **Transport Play** | \x01! |
| **Transport Record** | \x02" |
| **Transport Rewind** | @ |
| **Transport Stop** | \x00 |